Precept 1

Goals

- Review the concepts of sample space, outcome, events, and probability measures.
- Apply the axioms of probability to prove some properties.
- Illustrate how different modeling assumptions result in different probabilities for the same "question".

Exercises

Exercise 1 (Probability Rules)

Let $\mathbb{P}$ be a probability measure defined on $(\Omega, \mathcal{F})$. Prove the following properties:

1. $\mathbb{P}(\emptyset) = 0$
2. $\mathbb{P}(A \cap B^C) = \mathbb{P}(A) - \mathbb{P}(A \cap B)$
3. $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$

Exercise 2 (Assigning Probabilities to Events)

Suppose you have an unfair four-sided die (think triangular pyramid); where an even number has twice the probability of coming up as an odd number. You throw the die once and record the number on the face that's facing down.

1. What is the sample space of this random experiment?
2. Define the probability measure on this sample space.
3. Find the probability that you get a number less than 3.

Exercise 3 (Probability Paradox: The Importance of Precise Modeling)

1. *Cutting a cake:* It's Jessica's birthday and her friends throw her a surprise birthday party. Naturally, they bought her a chocolate cake with raspberry mousse, her favorite. She blows out the candles, and just can't wait to taste it, but unfortunately, her friends are quite absent-minded and they forgot to bring a suitable knife. Everyone goes searching, and for better or worse, that anime-loving friend of theirs brings out their katana sword. In the moment of excitement, and lack of judgment, they decide to play the following game: everyone gets

blindfolded and gets to take exactly one slice of the cake. The first person whose cut is larger than the radius of the cake wins and gets to eat the first slice. Naturally, Jessica goes first. What is the probability that she wins?

Mathematically this question essentially asks: *Given a fixed circle with radius r, what is the probability that a randomly selected chord has length greater than or equal to r?*

You might immediately notice that this problem exhibits the same kind of "imprecision" in this problem. There needs to be a fixed modeling assumption before we can go on with the problem. The core issue is: what does *randomly selected* mean? We need to fix a sample space and a probability measure; there are many reasonable ways that we can do this and we'll explore three possibilities now.

(a) **Independently and uniformly choose two points from the circumference of the circle to create a chord.**

(b) **Choosing the midpoint of the chord uniformly from the interior of the circle.**

(c) **Choosing the perpendicular distance of the chord to the center of the circle uniformly at random.** This one sounds a bit less intuitive. The sequence of steps is 1) choose a radius, 2) choose a random point on the radius (uniformly), and 3) draw the chord perpendicular at that point.

For each of the possibilities, answer the following questions:

- What is the sample space?
- How do we assign probabilities, i.e. what does "chosen uniformly" mean?

2. **Probability Paradox 2: Choose a number**

(a) A real number between 0 and 10 is chosen at random. What is the probability that it is greater than 5?

(b) A real number between 0 and 100 is chosen at random. What is the probability that it is greater than 25?

(c) What is the probability that the square of a random number from $[0, 10]$ is greater than 25?

(d) What is the probability that the square root of a random number from [0,100] is greater than 5?